

Bureau of Indian Standards

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IS 455:1989

1989

PORTLAND SLAG CEMENT - SPECIFICATION

SPECIFICATION

ICS 91.100.10

BIS 1989

Indian Standard

1 SCOPE

This standard covers the requirements for Portland Slag Cement (PSC) manufactured by inter-grinding Portland cement clinker with granulated slag and gypsum. PSC is particularly suitable for marine and mass concrete construction due to its low heat of hydration and sulfate resistance.

2 COMPOSITION

Portland Slag Cement shall contain: (a) Clinker: 25-50% by mass, (b) Granulated Slag: 40-70% by mass, (c) Gypsum: 3-5% by mass. The granulated slag shall conform to IS 455 Annex A.

3 REQUIREMENTS

Characteristic	33 Grade	43 Grade	53 Grade
Fineness (Blaine), m/kg, min.	300	300	300
Soundness (Le Chatelier), mm, max.	10	10	10
Initial Setting Time, min, min.	30	30	30
Final Setting Time, min, max.	600	600	600
Compressive Strength 3d, MPa, min.	8	18	23
Compressive Strength 7d, MPa, min.	14	28	33
Compressive Strength 28d, MPa, min.	33	43	53

4 SULFATE RESISTANCE

Portland Slag Cement exhibits excellent sulfate resistance due to the presence of slag. When tested as per IS 4031 (Part 12), the expansion at 6 months shall not exceed 0.05% in 5% sodium sulfate solution.

5 TEST METHODS

- Fineness: IS 4031 (Part 1) - Blaine Air Permeability
- Soundness: IS 4031 (Part 3) - Le Chatelier
- Setting Time: IS 4031 (Part 5) - Vicat Apparatus
- Compressive Strength: IS 4031 (Part 6) - 50mm Cubes
- Sulfate Resistance: IS 4031 (Part 12)